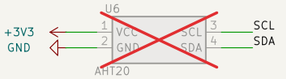
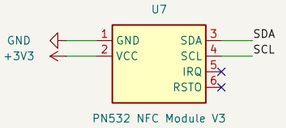
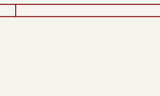


CSB and SDO use the module's own pull-ups: CSB high selects I2C; SDO picks address 0x76 or 0x77, and the firmware tries both.

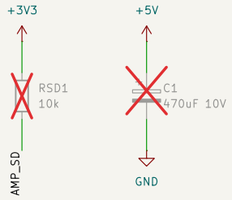


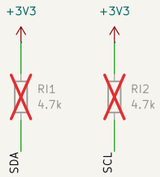


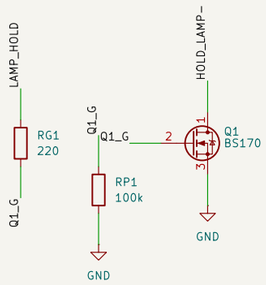




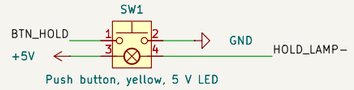


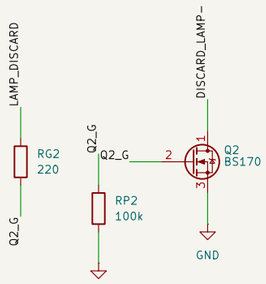




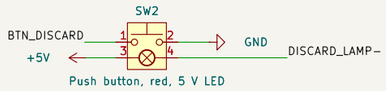












Intake panel, ESP32-C3. Net names on U1 GPIOs are the firmware pin constants without PIN_.
U2 OUT+ / OUT− are the module's screw terminal (J6): speaker LS1 red to +, black to −, nothing else.
Off the board, on harnesses: U3 (J1), U4 or U5 + U6 (J2), SW1 (J3), SW2 (J4), BZ1 (J5), LS1 (J6), U7 (J7).
U7: PN532 reader (Elechouse NFC Module V3, I2C mode: switch 1 ON, switch 2 OFF), fitted but optional; IRQ and RSTO unconnected.
I2C addresses: 0x3C U3, 0x76/0x77 U4 or U5, 0x38 U6, 0x24 U7.
Room air: fit U4 (BMP280 on a GY-BME/P280 board: temperature and pressure) or U5 + U6 (one BMP280 + AHT20 combination board on one 4-pin header: U5 is drawn as the combination board and U6 lists the AHT20's own ratings; adds humidity), not both.
DNP by default: RI1, RI2 (only if no I2C board has pull-ups), RSD1 (only if SD reads near 0 V).
C1 (only if 5 V sags on speech peaks). U2 GAIN open = 9 dB. GPIO 2, 8, 9 are strapping pins: unused.
Not drawn: BRD1 ElectroCookie full board; XU1, two 8-pin female headers socketing U1; F1, F2 speaker ferrules.
SW1, SW2: the 5 V LED built into each button has its own series resistor; LAMP in net names means that LED.
Q1, Q2: a 2N7000 may replace a BS170; its TO-92 pins are mirrored (S G D against D G S), so fit it turned round.

Sheet: /	
File: intake-panel.kicad_sch	
Title: Intake panel	
Size: A3	Date:
KiCad E.D.A. 10.0.5	Id: 1/1